

Homework 5: Cluster Analysis

S&DS 5360 | Multivariate Statistics

Brandon Tran (brandon.tran@yale.edu)

March 28, 2026

Setup & Data Overview

We continue our work with the Healthy Minds Survey (HMS) data. The variables are continuous, but lie on different scales due to the construction of each variable (a sum of underlying Likert-scale questions). Our total number of observations is $n = 68,062$.

- Flourishing (`flourish`): 8–56
- Depression (`deprawsc`): 0–27
- Anxiety (`anx_score`): 0–21
- Loneliness (`lonesc`): 3–9
- Disordered Eating (`ed_sde`): 0–5

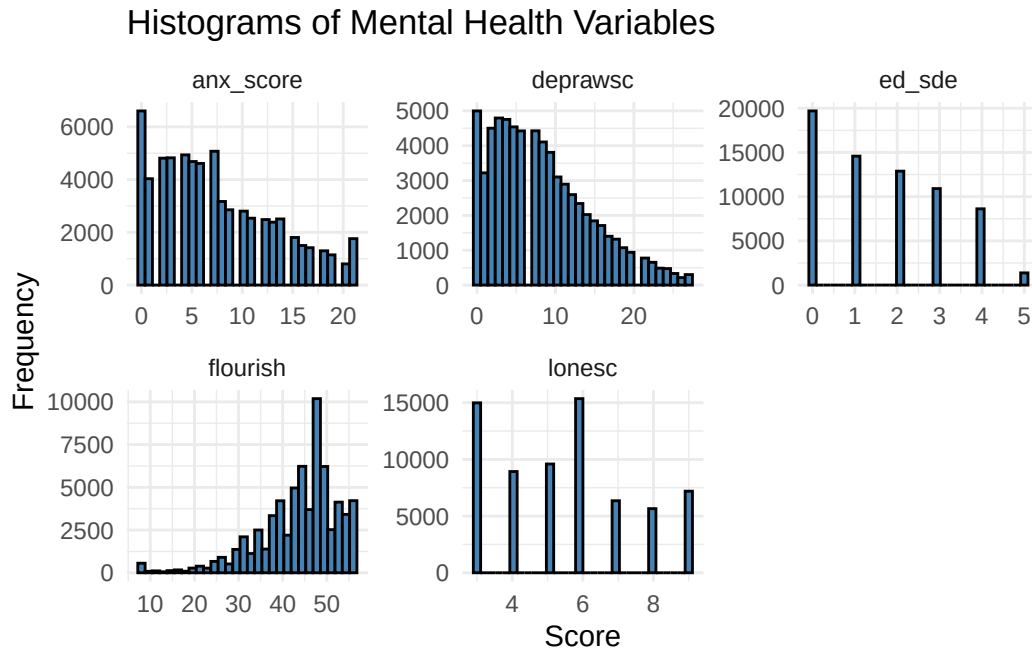
We omit the code to preprocess the data.

Part 1 | Distance Metrics, Standardization, and Transformation

Because we are clustering strictly on continuous, numeric mental health summary scores, standard geometric distance metrics are the most appropriate for our data. As such, we will elect to use the traditional Euclidean distance (straight-line distance between students' standardized score profiles). As a secondary metric, we elect to use Manhattan distance, which is less sensitive to extreme outliers.

First, recalling our previous analyses on this dataset, we may need to transform our variables to normalize our data.

```
df %>%
  pivot_longer(cols = everything(), names_to = "Variable", values_to = "Score") %>%
  ggplot(aes(x = Score)) +
  geom_histogram(bins = 30, fill = "steelblue", color = "black") +
  facet_wrap(~ Variable, scales = "free") +
  theme_minimal() +
  labs(title = "Histograms of Mental Health Variables",
       x = "Score", y = "Frequency")
```



The histograms for anxiety, depression, and disordered eating all exhibit right skewness. Flourish exhibits a left skew, and loneliness retains the underlying ordinal structure from its smaller subset of questions. To correct these, we propose a log transformation to the right-skewed variables and a square transformation to the left-skewed variable. We present the resulting Q-Q plots following the transformations:

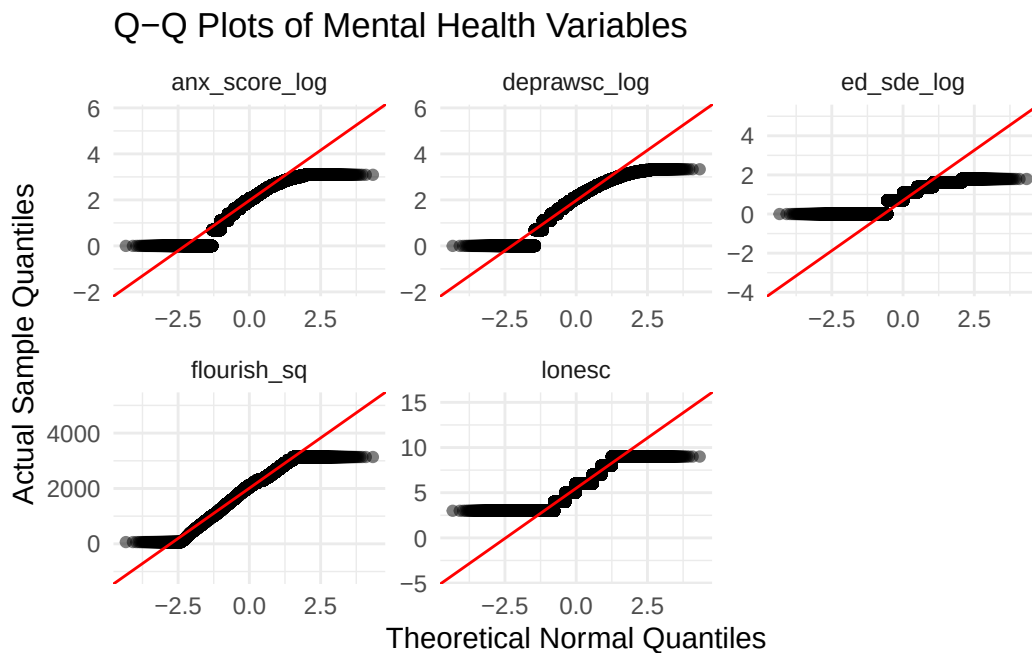
```
df_transformed <- df %>%
  mutate(
    deprawsc_log = log(deprawsc + 1),
    anx_score_log = log(anx_score + 1),
    ed_sde_log = log(ed_sde + 1),
    flourish_sq = flourish^2) %>%
  select(-deprawsc, -anx_score, -ed_sde, -flourish)
```

```

par(mfrow = c(2,3))

df_transformed %>%
  pivot_longer(cols = everything(), names_to = "Variable", values_to = "Score") %>%
  ggplot(aes(sample = Score)) +
  stat_qq(color = "black", alpha = 0.5) +
  stat_qq_line(color = "red") +
  facet_wrap(~ Variable, scales = "free") +
  theme_minimal() +
  labs(title = "Q-Q Plots of Mental Health Variables",
       x = "Theoretical Normal Quantiles",
       y = "Actual Sample Quantiles")

```



We see a staircase effect in the Q-Q plots, which is inherent to the bounded survey data we are working with; transformations cannot change this. However, a bulk of our observations follow the Q-Q line, and observe that our post-log transformed variables exhibit lighter tails, meaning our outliers were pulled inward. This ensures that extreme outliers will not disproportionately inflate the Euclidean distance calculations between students. We see a similar effect with our squared `flourish` variable.

As can be observed in the overview in the previous section, each continuous variable exists along a different scale. We will need to standardize our variables; else, flourishing and other

variables with large ranges will dominate. To standardize, we convert our scores to Z-scores ($Z \sim N(0, 1)$).

```
df_scaled <- scale(df_transformed)
```

Part 2 | Hierarchical Cluster Analysis

With our data transformed and scaled, we now perform hierarchical cluster analysis.